

Deep Assessment Report

We are a US-based corporate issuer with an existing, recurring commercial paper (CP) program used for short-term...

A case-specific assessment report for tokenization viability and value creation.

September 13, 2026

A TVAC assessment by Astoval

CLARITY
BEFORE
COMMITMENT.

About this report

TVAC evaluates whether tokenizing a specific asset / structure creates net added value, using the five-factor model:

Deep Assessment Report	We are a US-based corporate issuer with an existing, recurring commercial paper (CP) program used for short-term...
Generated	September 13, 2026
Methodology	TVAC — Tokenization Value-Add Calculator
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$$\text{Added Value} = (\text{New Opportunities (NO)} + \text{Cost Savings (CS)} + \text{Risk Reduction (RR)}) - (\text{Tokenization Costs (TC)} + \text{New Risks (NR)})$$

This document is generated using the TVAC (Tokenization Value-Add Calculator) methodology and provides methodology-based informational analysis and decision-support only. It is not legal, regulatory, financial, investment, or tax advice.

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01 | Executive summary

TOKENIZATION VALUE VERDICT

Viabile

ADDED VALUE

+8.00

plus 15.00 minus 7.00

Headline: The analysis supports a viable case for proceeding with tokenization under the proposed direction. The assessment lands at Viable with an estimated Net Added Value of 8.00 (Plus: 15.00 · Minus: 7.00).

Important interpretation note: This verdict concerns tokenization added value only. A Viable or Conditional Go verdict does not mean that the underlying asset is a good investment, that projected returns are realistic, that credit risk is acceptable, or that the instrument is suitable for any investor.

Yield neutrality note: The stated coupon, yield, interest rate, or expected return has not been treated as tokenization value in itself. TVAC does not assess whether projected returns are realistic, sustainable, appropriately priced for risk, or suitable for investors. Yield may be relevant to investor demand, but it does not by itself demonstrate that tokenization adds value.

Case design stage: Structured Concept

This assessment is based on a partially defined design. The case is sufficiently clear for a meaningful case-specific evaluation, but several structural choices remain open.

KEY TAKEAWAYS

- 01 Key findings:** The strongest support comes from new opportunities, cost savings, and risk reduction. Across New Opportunities, Cost Savings and Risk Reduction, the combined Plus score is 15.00. The TVAC methodology indicates a positive value hypothesis under the stated design, but the benefits still require validation.
- 02 Key constraints:** The main discipline point is to keep tokenization costs and new risks contained at 7.00 and avoid scope creep that would dilute the current value case.
- 03 Decision posture:** Proceed with disciplined implementation. Preserve the features that create value, but avoid unnecessary expansion of jurisdictional scope, liquidity promises, or technical complexity beyond what the case actually needs.
- 04 Impact:** This is primarily a controlled progression case. The upside appears plausible under the stated design, but it must be validated and captured without reintroducing avoidable TC/NR through overengineering or distribution overreach.

1.1 What is different about this case

What is different about this case is that the issuer already has a conventional commercial-paper programme and is testing whether tokenization can improve issuance, settlement, position reporting and lifecycle administration. The value case therefore depends on measurable process improvement, authoritative records, DvP and recovery controls, and retirement of legacy steps — not on new capital access, anchor investors, market making or assumed secondary liquidity.

02 | Asset class & market context

This is primarily an issuance-and-lifecycle optimisation case within an established recurring debt-issuance programme. Tokenization value depends on whether the tokenized tranche can improve allocation, settlement, authoritative position records, reporting, interest or discount and maturity administration, exception handling and reuse across later issuances. It is not assessed as a new-capital-access or secondary-liquidity strategy.

Current market phase: Case-derived structural context only; external market maturity is not assessed from verified market sources for this case.

Typical structural strengths:

- The issuer has an established conventional issuance-programme baseline against which operational improvement can be tested.
- The stated pilot targets concrete issuance and servicing frictions: allocation, settlement, position reporting, interest or discount administration and maturity processing.
- The professional-investor and no-liquidity-thesis perimeter keeps the pilot focused on controlled operational value rather than speculative market expansion.

Key structural weaknesses & frictions:

- The authoritative record, custody or wallet-control model and responsibility split across issuer, arranger, registry operator and settlement bank remain unresolved.
- Cash-leg finality, failed-settlement handling, cancellation, rollback and recovery must be validated before settlement-sensitive benefits can be credited.
- A smaller tranche may struggle to absorb fixed setup costs, and value weakens materially if legacy issuance and servicing steps remain permanently in parallel.

03 | Analysis result

ADDED VALUE

+8.00

PLUS

15.00

MINUS

7.00

Verdict rationale:

This case is assessed as Viable because the current user-defined structure produces Added Value +8 (plus 15 minus 7), which places the case in the Viable band under the TVAC methodology. In practical terms, the strongest supportive driver is New Opportunities (5/10), while the main friction is Tokenization Costs (5/10). For this recurring US commercial paper program, the main opportunity is not new investor reach but a token-enabled operating model that can unify position records and automate repeat lifecycle events. Even in a conservative design, a tokenized commercial paper tranche is likely to require platform, custody, integration, governance, and documentation work beyond the conventional process. This rationale concerns tokenization added value only and does not validate legal, regulatory, tax, financial, investment, credit, yield, issuer, or implementation suitability.

Factor score profile

Factor	Score	Case-specific driver
New Opportunities	5/10	For this recurring US commercial paper program, the main opportunity is not new investor reach but a token-enabled operating model that can unify position records and automate repeat lifecycle events.
Cost Savings	5/10	The user states that manual confirmations, repeated reconciliations, and exception handling are current pain points, so a tokenized tranche could reduce operating effort where those steps are actually replaced.
Risk Reduction	5/10	The described use of a single shared position record could reduce reconciliation breaks and improve auditability across the issuer treasury team, agents, custodians, and investors.
Tokenization Costs	5/10	Even in a conservative design, a tokenized commercial paper tranche is likely to require platform, custody, integration, governance, and documentation work beyond the conventional process.
New Risks	2/10	New risks exist around legal-to-technical mapping, provider dependency, and the relationship between the token record, custody record, and program documentation.

Added Value: 8.00 (plus 15.00 / minus 7.00)

Highest-priority evidence gaps:

- No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.

- Obtain written legal and control analysis of the token layer, authoritative records, participant and intermediary roles, custody or collateral control, settlement finality and the operating perimeter.
- Document and validate the relationship between the tokenized workflow, authoritative custody or agent records, collateral-control rights, asset- and cash-leg finality, exception handling, rollback and recovery.

Regulatory coverage: United States · Unverified

This report uses approved baseline context and any activated versioned claims. No verified case-specific jurisdictional claim is activated unless explicitly shown below; final local application remains unresolved and requires qualified local verification.

Claim IDs, source versions and validation questions are documented in Appendix A.6.

3.1 Value pathways

This section isolates the mechanisms through which tokenization could create incremental value over a realistic conventional alternative. It is not an assessment of whether the underlying asset or financing need is attractive in its own right.

- The core value logic is operational optimization of a recurring US commercial paper program, not expansion into retail distribution or open-market liquidity.
- The user describes repeated reconciliations across the treasury team, issuing or placing agents, custodians, and investors; moving the tokenized tranche to a single shared position record could reduce those recurring breaks.
- Automating issuance confirmations, coupon or accrual reporting, and maturity repayment notices could improve repeatability across short-dated issuance cycles, provided the tokenized tranche does not keep duplicate legacy steps alive.
- DvP-style workflows are relevant because the current process reportedly suffers from operational friction and occasional settlement fails, but the stated cash leg remains on T+1/2 rails, so the case does not depend on instant settlement.
- Because secondary-market liquidity is explicitly not a target, the business case stands or falls mainly on issuer-side control, administration, and settlement discipline.

3.2 Investor segmentation

Investor scope affects both upside and friction. A narrower professional or institutional audience can reduce distribution and compliance complexity, but it can also limit the access or liquidity benefit that tokenization is expected to create.

- The target audience is limited to professional and accredited investors, with retail investors explicitly excluded.
- That segmentation supports the case's control-oriented design because the objective is not mass-market access but a disciplined tokenized tranche inside an existing working-capital program.
- The input does not state whether buyers are expected to come primarily from the issuer's existing CP investor base or from newly onboarded eligible investors, so expansion assumptions should remain limited.
- Whitelist-based participation and issuer or agent-approved transfers can support compliance discipline, but they also reduce transfer flexibility and

3.4 Regulatory path

This is regulatory scoping, not a legal opinion. The purpose is to identify the verified regulatory context and the design questions that materially affect the case, while making clear what still requires qualified jurisdiction-specific validation.

- TVAC has approved baseline regulatory context for United States, but it has not activated a verified case-specific local claim. The frameworks below are used for structured scoping only; applicability and current detailed conditions remain to be independently confirmed.
- At scoping level, a token representing debt, equity, fund interests or another investment exposure should be analysed under the U.S. federal securities-law framework. Tokenization does not itself determine the instrument's legal classification.
- For U.S. securities capital formation, Regulation D, Regulation A and registered offerings are distinct pathways. Regulation S may be relevant to offshore offers and sales. TVAC may compare their possible relevance, but must not conclude that the final case qualifies for any pathway.
- Broker-dealer, alternative trading system, transfer-agent, custody and recordkeeping questions may arise according to the functions actually performed, the instrument classification and the proposed secondary-transfer model. A technology label does not decide those roles.

3.5 Custody & venue fit

Custody and venue design should be proportionate to the value thesis. If the case does not depend on continuous secondary trading, extra custody, venue or transfer complexity should be justified by a concrete operational, control or distribution benefit.

- A regulated-custodian model fits the user's institutional control objective better than a self-custody or open-access model would for this US commercial paper program.
- An OTC venue posture with no targeted secondary-market liquidity is consistent with the stated objective of post-trade efficiency rather than liquidity creation.
- The transfer restriction design—whitelist plus accredited or professional-only plus issuer or agent-approved transfers—supports control, but the legal enforceability of those controls should be validated separately from their technical implementation.
- Because the settlement and cash models are both stated as T+1/2, the architecture appears proportionate to an operational-efficiency thesis, but it

therefore do not create a liquidity story by themselves.

- For this commercial paper use case, investor segmentation matters mainly because it shapes onboarding, transfer controls, and intermediary responsibilities rather than retail distribution economics.

3.3 Operational costs

This section distinguishes plausible process savings from the additional cost of introducing the token layer. A saving becomes a strong value driver only when it can be compared with the current process and is not offset by data, integration, provider or compliance work.

- The recurring nature of the commercial paper program makes repeated workflow reuse plausible, which is one reason the user expects measurable savings from fewer manual confirmations and reconciliation breaks.
- Those savings are more plausible here because the user explicitly states that duplicated legacy steps for the tokenized tranche would be decommissioned rather than maintained indefinitely.
- At the same time, tokenization is still likely to require platform, custody, integration, governance, and documentation costs, and no binding provider pricing has been supplied.
- No validated conventional baseline was provided for current per-issuance operating cost, exception handling cost, or settlement-fail cost, so TVAC assesses the efficiency case as plausible but not yet measured.
- The case therefore supports meaningful operational-efficiency potential without proving realized savings at this stage.

should not be presented as evidence of instant or fully atomic settlement.

- Custody and venue fit are therefore directionally sound for a conservative tokenized CP tranche, while authoritative records and intermediary-role allocation remain unresolved.

3.6 Settlement & Cash-Leg Readiness

Settlement is assessed against the case's actual value logic, not against an assumption that tokenization requires digital money or T+0. The key question is whether the cash leg can operate reliably for the intended investors and operating model without unnecessary complexity.

- Cash-leg dependency is assessed as Critical. In practical terms, the settlement and cash infrastructure is central to the claimed value proposition and must be treated as core infrastructure. The assessment is based on the stated instrument/asset type (wrapper) and the stated liquidity intent (none), not on a generic assumption that every tokenized asset requires advanced digital money.
- Current payment / settlement indication: T+1/T+2 securities settlement. Readiness is assessed as Defined conventional design.
- High dependency: only user-stated benefits that depend on liquidity, DvP, collateral mobility, automated payments or 24/7 settlement should be constrained by weak cash-leg evidence.
- Design implication: if the core value proposition depends on atomic DvP, 24/7 liquidity, collateral mobility or digital cash settlement, the cash leg must be treated as core infrastructure rather than a late operational detail.

04 | Stakeholder added value

The stakeholder view shows where this case creates value and where its costs and risks actually sit. The scores below do not change the overall TVAC score; they expose whether the incentives are aligned across the parties who must make the structure work.

4.1 Stakeholder added value scores

Stakeholder	NO	CS	RR	TC	NR	Net
Issuer / programme sponsor Expected value depends mainly on whether the tokenized tranche improves issuance, settlement, position reporting and lifecycle administration relative to the issuer's established recurring issuance programme, without creating a permanent parallel process.	5.0	5.0	4.0	9.0	5.0	0.00
Professional investors Investor value depends mainly on clear rights, reliable position records, controlled settlement and accurate lifecycle servicing rather than on promised liquidity or expanded access.	5.0	3.0	6.0	5.0	5.0	4.00
Issuance, registry, custody and settlement providers Provider value depends on delivering a coherent issuance and lifecycle stack that integrates with the issuer's existing systems and removes measurable friction rather than merely adding another technology layer.	6.0	2.0	2.0	4.0	1.0	5.00

Here, overall project TC can remain moderate while Asset Owner TC is materially higher, because setup, integration, legal/compliance work, and delivery burden are concentrated on the initiating party rather than spread evenly across the structure. Likewise, stakeholder NR can exceed overall project NR where a specific party bears more custody, transfer, disclosure, or exit-path uncertainty than the structure as a whole.

Time-horizon interpretation: For this high-volume institutional workflow, the initiating institution's stakeholder score should be read primarily as an implementation-phase view. The setup burden is front-loaded. The intended economic case depends on cumulative run-state savings and risk reduction over a multi-year operating horizon once the stated legacy reconciliation and processing steps are actually retired. TVAC does not calculate ROI or a payback period; this interpretation is conditional on measurable evidence that the stated benefits are captured in practice.

4.2 Issuer / programme sponsor

Expected value (plus)

- Potential value comes from fewer manual handoffs, cleaner allocation and settlement, better position visibility and more controlled interest, maturity and lifecycle administration.
- A successful pilot can create reusable operating capability for later issuances without relying on broader investor access or secondary-market liquidity.

Costs & risks carried (minus)

- The issuer still carries legal-structure, integration, provider, governance and change-management costs before operational gains are demonstrated.
- Value erodes if the tokenized tranche remains dependent on duplicate records, manual approvals or permanent parallel servicing.

What to agree / do next

- Define the authoritative record, responsibility split, lifecycle event model and evidence required to retire legacy steps.
- Compare the full tokenized-tranche cost and operating performance with the existing issuance-programme baseline before scaling.

4.3 Professional investors

Expected value (plus)

- Cleaner confirmations, position reporting and lifecycle communications may improve the holding and servicing experience.
- A validated DvP process can reduce settlement uncertainty without changing the conventional economic terms of the instrument.

Costs & risks carried (minus)

- Investors need clarity on the authoritative record, custody or wallet control, transfer restrictions, failed-settlement handling and fallback arrangements.
- A new token layer adds provider and operational dependencies even where no public secondary market is planned.

What to agree / do next

- Require a clear rights-and-operations note covering record authority, settlement finality, custody, transfers, interest or discount, maturity processing and incident recovery.
- Do not treat the pilot as offering additional liquidity unless a separate, validated transfer pathway is actually introduced.

4.4 Issuance, registry, custody and settlement providers

Expected value (plus)

- Providers can create durable value through reliable registry integration, controlled DvP, lifecycle automation, audit evidence and reusable issuance workflows.
- A repeat issuer offers a credible path to reuse if the first tranche proves operationally superior to the conventional process.

Costs & risks carried (minus)

- Responsibility gaps between arranger, registry or transfer operator, custodian, wallet provider and settlement bank can create high error and recovery costs.
- Commercial value is weak if fees and integration burden are not tied to measurable process retirement and service outcomes.

What to agree / do next

- Agree a detailed RACI, authoritative-record model, DvP sequence, exception authority, recovery process, service levels and exit or portability provisions.
- Tie provider commitments and scale-up to evidenced reductions in manual work, reconciliation and lifecycle-processing burden.

4.5 Who bears the costs and risks?

For this qualified high-volume institutional workflow, the initiating institution may carry most front-loaded integration and governance costs. That does not by itself mean the economics favour service providers. The relevant test is whether the initiating institution captures measurable cumulative savings and risk reduction once the workflow reaches run-state and legacy steps are actually retired.

- A neutral or low near-term initiating-institution score should trigger an evidence and cost-allocation review, not an automatic rejection. It becomes a warning sign where the tokenized lane leaves legacy processes in parallel, the expected operational benefits are not demonstrated, or provider fees absorb the realised gains.
- Eligible counterparties should still receive clear evidence of operational controls, custody synchronisation, settlement finality and the allocation of exception-handling costs.

05 | Factor breakdown – detailed explanation

This chapter explains the causal basis of the five scores: which facts, assumptions and evidence gaps create upside, and which ones introduce additional friction. The three positive factors ask what tokenization adds relative to a realistic conventional alternative; the two negative factors ask what extra cost or risk the token layer introduces. A low positive score therefore means limited or weakly evidenced tokenization benefit, while a low negative score means limited additional friction. The scores are calibrated methodology judgments, not measured outcomes or legal conclusions.

Overall, this structure creates a clearly positive balance: strong upside from new opportunities and/or risk reduction outweighs tokenization costs and new risks, provided the project is executed with solid governance and implementation discipline. With an Added Value score of 8, the value case is clearly positive rather than marginal. Net Added Value is primarily driven by new opportunities at 5/10 on the upside, and constrained mainly by tokenization costs at 5/10 on the friction side.

New opportunities (NO)

- For this recurring US commercial paper program, the main opportunity is not new investor reach but a token-enabled operating model that can unify position records and automate repeat lifecycle events.
- That opportunity is difficult to replicate fully with fragmented bilateral confirmations because the user specifically targets shared records, whitelist controls, and structured lifecycle messaging on the tokenized tranche.
- Upside is deliberately capped by the stated absence of open secondary liquidity, the United States-only perimeter, and the fact that the note is described as economically equivalent to the existing CP rather than as a product with materially expanded rights.

Cost savings (CS)

- The user states that manual confirmations, repeated reconciliations, and exception handling are current pain points, so a tokenized tranche could reduce operating effort where those steps are actually replaced.
- The stated commitment to decommission duplicated legacy steps for the tokenized tranche is important because it turns tokenization from an overlay into a potential process redesign.
- Savings are still assessed conservatively because no quantified baseline, provider pricing, or realized operating data were supplied.

Risk reduction (RR)

- The described use of a single shared position record could reduce reconciliation breaks and improve auditability across the issuer treasury team, agents, custodians, and investors.
- DvP-style settlement workflows are directly relevant because the current CP process reportedly experiences operational friction and occasional settlement fails.

- Whitelist-based participation, institutional custody, and controlled transfers support a lower-risk operating perimeter than an open-access or retail design would.

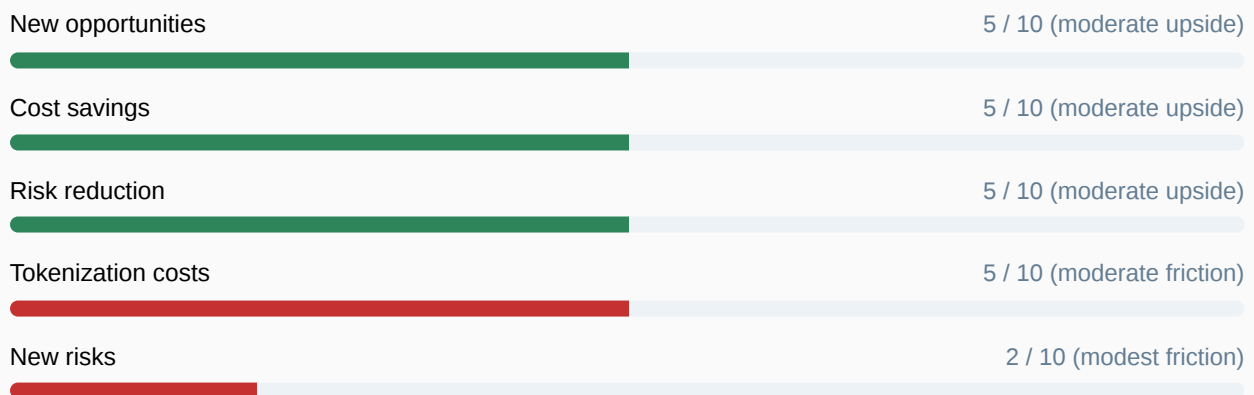
Tokenization costs (TC)

- Even in a conservative design, a tokenized commercial paper tranche is likely to require platform, custody, integration, governance, and documentation work beyond the conventional process.
- The issuer still needs coordination across existing treasury operations, agents, custodians, and the token stack, so tokenization does not remove all operational complexity.
- Because no binding commercial terms were provided, TVAC treats the cost burden as meaningful but not clearly excessive for a recurring institutional program.

New risks (NR)

- New risks exist around legal-to-technical mapping, provider dependency, and the relationship between the token record, custody record, and program documentation.
- Those risks are moderated by the narrow US-only, professional or accredited, whitelisted design and by the fact that the case does not rely on retail participation or open-market liquidity.
- The limited-liquidity posture and regulated-custodian model keep the risk build-up lower than it would be in a broader or more market-facing tokenization design.

Factor scores (0–10). Green bars indicate upside factors (New opportunities, Cost savings, Risk reduction). Red bars indicate friction/downside (Tokenization costs, New risks). The text next to each score gives a qualitative label so the numbers are less abstract.



Plus (NO + CS + RR): 15

Minus (TC + NR): 7

Added Value (Plus – Minus): 8

06 | Key risks & mitigation

This matrix prioritizes the report's identified risks by impact and likelihood. It is not a measured probability model; each position should be read together with the case-specific description and mitigation.

		IMPACT				
		1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 High	5 High					
	4 Med-high		R2 R5	R1 R3		
	3 Medium		R4 R6			
	2 Med-low					
	1 Low					
		LIKELIHOOD				

Risk zones are indicative prioritization bands, not measured probabilities.

PRIORITY RISKS

R1 Misaligned incentives / vendor-capture risk

Impact 4 · Likelihood 3

Mitigation: Use stage-gating based on legal/control validation and measurable operational-readiness evidence; link provider fees and scale-up to issuance completion, process retirement, service levels, reconciliation performance and lifecycle-delivery outcomes; preserve portability and exit rights if the provider arrangement changes.

R2 Authoritative record ambiguity

Impact 4 · Likelihood 2

Mitigation: Document which record governs ownership, transfers, and payment obligations across the token record, custodian books, and existing CP documentation.

R3 Parallel-process benefit leakage

Impact 4 · Likelihood 3

Mitigation: Make decommissioning of duplicated tokenized-tranche legacy steps an explicit operating requirement with exception-only fallbacks.

R4 DvP-style workflow underperformance

Impact 3 · Likelihood 2

Mitigation: Test end-to-end settlement workflows, cash-leg dependencies, and exception handling before relying on settlement-fail reduction assumptions.

R5 Intermediary-role misalignment

Impact 4 · Likelihood 2

Mitigation: Obtain independent US analysis of broker-dealer, custody, transfer-agent, and OTC transfer functions created by the final operating model.

R6 **Provider dependency in a recurring issuance process**

Impact 3 · Likelihood 2

Mitigation: Secure binding scope, service-level, pricing, portability, and exit terms for the custodian and platform stack.

HOW TO READ THIS PAGE

Higher placement indicates greater impact; movement to the right indicates greater likelihood. Risks that share a cell can still have different causes and require different responses.

07 | Recommendations & next steps

7.1 Design priorities and decision gates

This is an issuance and lifecycle-optimisation case for a conventional recurring debt-issuance programme, not a capital-access or secondary-liquidity case. The priorities focus on proving operational improvement against the existing issuance baseline.

7.1.1 Prove improvement against the existing issuance-programme baseline

Measure manual handoffs, allocation and settlement effort, reconciliation, position reporting, interest or discount administration, maturity processing, exceptions and lifecycle-processing cost before the pilot. Scale only where the tokenized tranche produces a durable improvement over the established process.

7.1.2 Resolve authoritative records, responsibility and DvP recovery

Define the authoritative instrument and investor record, the role of the token layer, the responsibility split across issuer, arranger, registry or transfer operator, custodian, wallet provider and settlement bank, and the rules for finality, cancellation, reversal, rollback and recovery.

7.1.3 Retire legacy steps and align provider economics with evidence

Specify which legacy issuance and servicing steps may be removed after validation. Tie provider commitments, fees and scale-up to measurable process retirement, control evidence, service levels, portability and incident performance.

7.1.4 Execution implication for an issuance and lifecycle pilot

Proceed through a bounded professional-investor pilot only if the operating model, authoritative record, DvP and cash-leg controls, lifecycle servicing and recovery arrangements are independently validated. Do not add anchor-investor, fundraising, market-making or liquidity objectives unless the case is separately redesigned and reassessed.

7.2 Design what-if scenarios

These scenarios test the operational value of the tokenized tranche relative to the issuer's existing issuance programme; they do not assume new capital access or secondary liquidity.

- The pilot removes defined manual and reconciliation steps while preserving authoritative records and controls: Cost Savings and Risk Reduction become more durable.
- The token layer remains parallel to existing records and servicing: Tokenization Costs rise and the operational value case weakens.

- Cash-leg finality, failed-settlement recovery or responsibility allocation remains unresolved: New Risks should be treated as higher and scale-up should pause.

7.3 Suggested next steps

- Establish the conventional issuance and servicing baseline and pilot success thresholds.
- Document the authoritative record, token role, RACI, lifecycle event model and recordkeeping obligations.
- Validate DvP, cash-leg finality, failed-settlement, cancellation, rollback and recovery procedures end to end.
- Agree which legacy steps can be retired and tie provider economics and scale-up to measured evidence.

Appendices

- Appendix A — Method & interpretation
- Appendix B — TVAC Provider Directory (illustrative)
- Appendix C — Glossary & abbreviations
- Appendix D — User-supplied case inputs
- Appendix E — Disclaimer

Appendix A — Method & interpretation

A.1 How to read this report

HOW TO READ THIS REPORT This report is a structured decision-support assessment of the specific tokenization case described in the inputs. It is designed for senior stakeholders who need an indicative Tokenization Value Verdict, the reasons behind it, and practical design levers. How to use the report (recommended sequence)

1. Start with the Tokenization Value Verdict (Viable / Conditional Go / Non-viable) and the top drivers.
 2. Read “Decision Integrity & Limits” to understand scope, boundaries, and what would change the verdict.
 3. Use the analysis sections to trace the conclusion across opportunity, cost, risk, and regulatory context (indicative).
 4. Use “Sensitivity & Decision Levers” and the recommendations as a design tool: adjust the few variables that move outcomes most.
- Recommendations in this report are analytical suggestions derived from the TVAC methodology and do not constitute legal, regulatory, financial, tax, investment, or implementation advice. Reading discipline
 - Treat uncertain inputs as hypotheses that must be verified.
 - Prefer staged commitments: validate the highest-impact uncertainties before committing to full implementation.

A.2 Decision integrity & interpretation

DECISION INTEGRITY & LIMITS This report provides informational decision support only. It does not constitute legal, regulatory, tax, financial, or investment advice, and it does not provide a compliance determination, valuation, or provider endorsement. Case snapshot (as stated)

- Issuance jurisdiction: United States
- Sale jurisdictions / target markets: United States
- Asset / instrument type (indicative): wrapper
- Liquidity intent (as stated): none
- Payment / cash-leg indication: T+1/T+2 securities settlement
- Cash-leg dependency / readiness: Critical dependency / Defined conventional design readiness What the Tokenization Value Verdict means
- Viable: Tokenization added value is plausible under the stated assumptions, subject to normal verification and execution.
- Conditional Go: Tokenization added value is design-dependent. Proceeding without meeting the listed conditions materially increases execution risk.
- Non-viable: Under current assumptions, tokenization is unlikely to outperform a conventional structure on a risk-adjusted basis. What can change the verdict (high impact)
- Distribution scope and investor segmentation (jurisdiction count, retail vs professional posture)
- Rights mapping and structural clarity (asset/SPV structure, cash-flow logic, investor protections)
- Custody / controls / transfer restrictions (institutional-grade requirements vs complexity)
- Liquidity assumptions (timing, venue feasibility, realistic market depth)
- Payment / cash-leg readiness where the case depends on DvP, digital cash, 24/7 settlement, collateral mobility or automated lifecycle payments
- Tokenization cost profile (legal/compliance/ops overhead and non-scalable components)
- Regulatory clarity and constraints in the stated jurisdictions (indicative; verify with qualified advisers) What TVAC did not assess
- No definitive legal, regulatory, tax, or investment conclusions
- No asset valuation, yield projections, or market-price forecasts
- No assurance of secondary-market liquidity or investor demand
- No operational readiness audit beyond the indicative checklist

A.3 Minimum evidence checklist

MINIMUM EVIDENCE & READINESS CHECKLIST (indicative) Before proceeding toward implementation, decision-makers should be able to substantiate a minimum level of evidence across the areas below. This checklist is not exhaustive and does not imply regulatory sufficiency. It reflects common governance expectations in institutional-grade projects. Highest-priority gaps in this case (derived from inputs/completeness)

1. No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
 2. Obtain written legal and control analysis of the token layer, authoritative records, participant and intermediary roles, custody or collateral control, settlement finality and the operating perimeter.
 3. Document and validate the relationship between the tokenized workflow, authoritative custody or agent records, collateral-control rights, asset- and cash-leg finality, exception handling, rollback and recovery. Structural & economic foundations
- Clear description of the underlying asset and ownership structure
 - Defined economic rights and cash-flow logic
 - Draft term sheet or structural outline
 - Clear rationale for tokenization vs alternatives Regulatory & distribution posture
 - Issuance jurisdiction clearly defined
 - Explicit list of sale jurisdictions
 - Documented investor type assumptions
 - Initial regulatory scoping (high-level) Token & control design
 - High-level token design and rights mapping
 - Transfer restriction and whitelist logic (conceptual)
 - Corporate actions and investor reporting approach
 - Governance for smart-contract or protocol changes (if relevant) Infrastructure & counterparties
 - Custody model (conceptual)
 - Indicative partner shortlist (non-binding)
 - Issuance, registry, and lifecycle approach
 - Payment / cash-leg and settlement approach, proportionate to the case's actual value proposition
 - Early assessment of secondary trading (if any) Cost & operational realism
 - High-level CAPEX and OPEX view
 - Identification of major cost drivers
 - Clear internal ownership of ongoing operations
 - Understanding of scalable vs non-scalable costs Lack of clarity across several of these areas materially increases execution risk, even where the conceptual case appears attractive.

A.4 Sensitivity & decision levers

SENSITIVITY & DECISION LEVERS (indicative) Deterministic verdict (source of truth): Viable. Small, targeted design changes can materially shift the outcome. Top drivers in this case (derived from scoring)

- Upside driver: Issuance and lifecycle improvement (value depends on measurable reductions in manual work, reconciliation, reporting and servicing)
- Risk driver: New risks (most sensitive to authoritative-record mapping, cash-leg finality, recovery design and provider-role clarity)
- Cost driver: Tokenization costs (most sensitive to fixed pilot costs, integration burden, duplicate processes and provider scope) High-impact decision levers (practical)
- Legacy-process retirement: identify which allocation, reconciliation, reporting, coupon and maturity steps can be removed after validation.
- Authoritative record and controls: define the legally and operationally authoritative record, the role of the token layer and the responsibility split across the operating stack.
- Settlement and recovery: validate DvP sequencing, cash-leg finality, failed-settlement handling, cancellation, reversal, rollback and recovery.

- Provider economics: tie fees and scale-up to service levels, measurable process improvement, portability and incident performance.
- Reuse across later issuances: improve economics by reusing validated integration, governance and lifecycle capabilities rather than rebuilding them for each tranche. Interpretation rule

Improve the operating evidence and control model before adding distribution, fundraising, trading-support or liquidity objectives. Those objectives would materially change the case and require a separate assessment.

A.5 How we compute Added Value

Added Value = (New Opportunities (NO) + Cost Savings (CS) + Risk Reduction (RR)) - (Tokenization Costs (TC) + New Risks (NR))

Plus (NO + CS + RR) = 15

Minus (TC + NR) = 7

Added Value (Plus - Minus) = 8

A.6 Evidence, provenance & regulatory source register

This register shows evidence status, activated verified regulatory claims, official sources and generic interface labels. It supports auditability but does not constitute legal advice or a conclusion that any cited rule applies to the final facts.

Supporting evidence: Preliminary · Any supporting evidence mentioned in the case description is treated as user-reported and has not been independently reviewed or verified by TVAC.

Regulatory coverage: United States · Unverified · Basic coverage · last reviewed 2026-07-15

Legal-advice boundary: TVAC provides regulatory scoping and decision support only. It does not determine how the law applies to the final facts, provide a legal opinion, confirm compliance, or replace qualified local counsel.

Score provenance

Factor scores are rubric-based methodology judgments grounded in supplied inputs and explicit evidence gaps. They are not measurements, legal conclusions or due-diligence findings.

Factor	Score	Provenance	Input references
New Opportunities	5/10	TVAC methodology assessment	case_description, instrument_type, investor_types, sale_jurisdictions, secondary_market_liquidity, transfer_restrictions
Cost Savings	5/10	TVAC methodology assessment	case_description, instrument_type, custody_model, settlement_model, transfer_restrictions
Risk Reduction	5/10	TVAC methodology assessment	case_description, custody_model, transfer_restrictions, settlement_model, investor_types
Tokenization Costs	5/10	TVAC methodology assessment	case_description, issuance_jurisdiction, sale_jurisdictions, instrument_type, custody_model, secondary_market_venue, settlement_model, offering_disclosure
New Risks	2/10	TVAC methodology assessment	case_description, issuance_jurisdiction, sale_jurisdictions, custody_model, secondary_market_venue, transfer_restrictions, settlement_model, offering_disclosure, tax_considerations

Activated verified regulatory claims

The table records claim scope and source IDs; the full case-facing explanation appears in Chapter 3.5.

No verified jurisdiction-specific claim was activated for the stated case facts.

Jurisdiction-specific validation questions

- **United States:** Which record is legally authoritative for the debt instrument and investor positions, and what legal and operational role does the token layer perform relative to that record?
- **United States:** How are responsibilities divided among issuer, arranger, transfer or registry operator, custodian, wallet provider and settlement bank for issuance, position maintenance and lifecycle events?
- **United States:** How are asset-leg and cash-leg finality, DvP sequencing, failed settlement, cancellation, reversal, rollback and recovery defined for the pilot?
- **United States:** Which eligibility, onboarding, disclosure, communications, marketing and transfer-control requirements apply in each stated jurisdiction to the professional-investor perimeter?
- **United States:** How must interest or discount, principal, maturity, corporate-action, tax and recordkeeping events be reflected across the authoritative record, token layer and existing issuer systems?
- **United States:** Which legacy issuance and servicing steps may be retired after validation, and what evidence is required before parallel processing can be discontinued?

Verified official sources

- SEC — Exempt Offerings overview · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Regulation D Offerings · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Regulation A · U.S. Securities and Exchange Commission · checked 2026-07-15
- eCFR — Regulation S: offers and sales outside the United States · U.S. Securities and Exchange Commission / Electronic Code of Federal Regulations · checked 2026-07-15
- SEC — Guide to Broker-Dealer Registration · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Alternative Trading Systems · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Transfer Agents · U.S. Securities and Exchange Commission · checked 2026-07-15

Generic interface labels

Generic TVAC interface label only. It is not treated as a jurisdiction-specific legal classification or legal requirement.

Field	Value
instrument_type	wrapper
custody_model	regulated custodian
secondary_market_liquidity	no secondary liquidity planned
secondary_market_venue	controlled OTC transfers
settlement_model	T+1/T+2 securities settlement
offering_disclosure	exemption-based disclosure route
investor_types	professional, qualified/accredited (jurisdiction-dependent)

A.7 Operational roadmap (structured)

This detailed implementation roadmap is retained as a reference, but moved out of the main analysis because much of it describes a generic project sequence. Use it after the case-specific decision points and recommendations in the main report have been resolved.

A.7.1 Operational roadmap (structured)

The items below are indicative planning scaffolding. They are not a substitute for jurisdiction-specific legal, tax, financial, and operational due diligence, and they must be tailored to the selected asset type, wrapper structure, investor segment, custody model, venue, and settlement design.

A.7.2 Phase 1 — Preparation & feasibility

Case-specific focus for this phase: Primary issuance jurisdiction: United States. Initial distribution scope: United States. Initial investor focus: professional investors, accredited / qualified investors.

Objectives

- Clarify the underlying asset/instrument, economic rights, and proposed token wrapper.
- Define target investor segments (professional / eligible / retail, if applicable) and distribution scope.
- Establish internal alignment on objectives, success criteria, and risk appetite.

Key decisions

- Token rights model (equity-like, debt-like, fund-like, revenue share, participation note, etc.).
- Transfer restriction regime (whitelisting, holding periods, jurisdictional gating).
- Custody and settlement direction (account-based vs. wallet-based; T+0 vs. T+1+).

Deliverables (checklist)

- One-page case definition: asset, wrapper, jurisdiction(s), investor segment, and target size.
- Preliminary term sheet outline (rights, restrictions, fees, lifecycle events, disclosures).
- High-level implementation estimate (time bands, cost bands, key dependencies).
- Initial risk register (top 10–15 risks with owners and first mitigations).

Primary stakeholders

- Issuer/asset owner (business sponsor, finance, legal/compliance, operations/IT).
- External legal counsel (securities, funds, structuring), tax adviser, and (if relevant) auditor.
- Shortlisted tokenization platform/technology provider, regulated custodian, and potential venue(s).

Dependencies / gates

- Clear legal characterization of the instrument and its regulatory perimeter in the issuance jurisdiction.
- Agreement on must-have constraints (investor eligibility, reporting, custody, settlement, data handling).
- Internal approval to proceed to structured design (budget and governance gate).

Risk hotspots

- Unclear instrument classification or incompatible distribution intent (e.g., retail scope without a verified pathway).
- Mismatch between user-stated transfer expectations and the legally and operationally supportable process.
- Hidden operational burdens (cap-table/admin, reporting, corporate actions, tax documentation).

A.7.3 Phase 2 — Legal, tax & regulatory structuring

Case-specific focus for this phase: Core legal and tax analysis should start from United States. Distribution permissions must be validated for United States. Generic TVAC disclosure label selected in the input: Exempt. The applicable local legal category remains to be confirmed. Transfer-control design to validate: Whitelist + accredited/professional-only + issuer/agent-approved transfers.

Objectives

- Obtain independent legal validation of a legally supportable issuance and distribution structure for the intended jurisdiction(s).
- Define disclosure, governance, and ongoing compliance obligations for the lifecycle of the instrument.
- Obtain qualified tax analysis of the proposed wrapper and cash-flow mechanics for issuer and investors.

Key decisions

- Issuance vehicle choice (direct issuer vs. SPV; fund wrapper vs. corporate wrapper).
- Offering and documentation pathway verified for the stated jurisdiction(s); do not infer a legal category from interface labels.
- Investor onboarding model (KYC/AML responsibilities, reliance model, delegated vs. in-house).
- Choice of legal documentation stack verified by qualified local advisers.

Deliverables (checklist)

- Legal characterization memo and jurisdiction matrix (requirements by country/investor segment).
- Draft offering documentation and investor disclosures.
- Transfer restriction policy and rulebook (whitelist logic, jurisdictional gating, and any explicitly selected holding-period rules).
- Tax memo(s) covering withholding, reporting, and cross-border distribution considerations.
- Compliance operating model (roles, escalation paths, record-keeping, audit trail).

Primary stakeholders

- Issuer legal/compliance, external counsel, tax advisers.
- Custodian/compliance function (if regulated custody is used), transfer agent (if applicable).
- Any transfer provider or environment explicitly retained in scope and independently validated.

Dependencies / gates

- Independent legal confirmation of the offering pathway and distribution permissions across the stated target jurisdictions.
- Operational feasibility of investor onboarding and transfer controls (including data privacy constraints).
- Formal sign-off on documentation pack and compliance model.

Risk hotspots

- Jurisdiction conflicts (marketing rules, suitability requirements, local registration triggers).
- Under-scoped ongoing obligations (periodic reporting, investor communications, corporate actions).
- Tax frictions that erode the added value thesis (withholding, reporting, complexity costs).

A.7.4 Phase 3 — Build, integration & vendor onboarding

Case-specific focus for this phase: The current custody design indicates regulated custodian. Its precise legal and operational role remains to be confirmed. The current transfer environment indicates controlled OTC transfers; this does not establish a local legal venue classification. The current settlement design indicates T+1/T+2 securities settlement. Its operational and legal fit remains to be validated. Transfer-rule implementation must reflect: Whitelist + accredited/professional-only + issuer/agent-approved transfers.

Objectives

- Finalize the technology stack and integrate custody, settlement, onboarding, and reporting workflows.
- Implement transfer rules and lifecycle event handling (issuance, redemptions, distributions, corporate actions).
- Establish operational readiness (controls, monitoring, incident response, reconciliation).

Key decisions

- Token standard / ledger choice and compatibility with custody and venue requirements.
- Settlement model (atomic delivery-versus-payment vs. coordinated settlement through intermediaries).
- Data and reporting architecture (who stores what; audit trail; privacy boundaries).
- Outsourcing vs. in-house responsibility split (transfer agent, admin, reporting).

Deliverables (checklist)

- Final technical architecture and integration plan (custody, venue, payments, identity/KYC).
- Rule engine / whitelist implementation and test suite (including jurisdictional gates).
- Operational runbooks (reconciliation, incident response, corporate actions, investor support).

- UAT results and sign-offs (issuer, custodian, venue, admin/TA as relevant).
- Security review (key management model, access control, change management, monitoring).

Primary stakeholders

- Issuer operations/IT, tokenization platform/tech provider, regulated custodian where explicitly selected.
- Selected transfer-environment integration team, payment rails (if applicable), admin/TA providers.
- Compliance and risk functions for control design and approval.

Dependencies / gates

- Signed vendor contracts, SLAs, and responsibility matrix (RACI) across the stack.
- Completed integration testing and reconciliation proof (end-to-end lifecycle flows).
- Readiness gate: controls, monitoring, and support operations in place.

Risk hotspots

- Integration complexity and reconciliation gaps across custody/venue/admin systems.
- Rule enforcement drift (transfer restrictions not consistently applied across venues/wallets).
- Operational single points of failure (provider concentration, key management, support load).

A.7.5 Phase 4 — Issuance, distribution & launch

Case-specific focus for this phase: Any legally required launch permissions and onboarding constraints must be independently confirmed for: United States. User-selected generic investor segment for launch planning: professional investors, accredited / qualified investors. Local legal category mapping remains to be confirmed. The launch transfer environment currently indicates controlled OTC transfers; this does not establish a local legal venue classification. The current liquidity intent is no secondary liquidity planned. No market status, matching demand or liquidity support is inferred.

Objectives

- Execute issuance mechanics and investor onboarding under the independently verified offering pathway.
- Operationalize disclosures, subscriptions, allocations, and post-issuance record-keeping.
- Operationalize only the user-stated and legally validated transfer process, if any; no venue listing, market making or liquidity support is assumed.

Key decisions

- Distribution channel mix (direct, intermediated, platform marketplace, placement agents) as legally validated.
- Subscription mechanics and settlement timing (funding rails, DVP, cutoff times) as actually specified.
- Transfer model, if any: stated approvals, eligibility, timing and limitations; no cadence or support mechanism is assumed.

Deliverables (checklist)

- Final investor communication pack and distribution checklist.
- Subscription and allocation runbook; reconciliation and exception handling.
- Go-live monitoring plan and incident escalation paths.
- Documented transfer-process setup, if applicable, and evidence-based monitoring metrics.

Primary stakeholders

- Issuer distribution team, platform operator, onboarding/KYC provider.
- Custodian, settlement and transfer providers explicitly selected for the stated process.
- Legal/compliance for marketing controls and ongoing disclosure obligations.

Dependencies / gates

- Approved marketing/distribution materials and compliance sign-off.
- End-to-end dry run of subscription, allocation, and post-issuance reporting.
- Operational readiness gate: support, monitoring, and governance in place.

Risk hotspots

- Distribution frictions (slow onboarding, suitability issues, cross-border marketing constraints).

- Unsupported transfer or liquidity expectations relative to the actual process and evidence.
- Operational load spikes (investor support, reconciliation exceptions, reporting deadlines).

A.7.6 Phase 5 — Post-issuance operations & monitoring

Case-specific focus for this phase: Ongoing governance and evidence trails must remain defensible under United States. Post-issuance compliance must continue to support holders across: United States. Ongoing operations must reflect the regulated custodian design once its precise role has been independently confirmed. Post-launch monitoring must reflect the actual validated transfer process currently described as controlled OTC transfers.

Objectives

- Run ongoing compliance, reporting, and investor communications through the instrument's lifecycle.
- Operate lifecycle events (distributions, redemptions, corporate actions) with strong controls and auditability.
- Monitor provider performance, costs, and risk posture; adjust as regulations and market conditions evolve.

Key decisions

- Governance cadence (board/committee oversight, change approvals, vendor reviews).
- Reporting scope and frequency (investor reporting, regulatory reporting, audit readiness).
- Contingency planning (provider exits, custody changes, venue strategy shifts).

Deliverables (checklist)

- Ongoing reporting calendar and control framework (incl. audits and reconciliations).
- Lifecycle event runbooks and evidence retention procedures.
- Vendor performance KPIs and incident log; periodic risk reviews.
- Contingency and change-management playbook (custody/venue migrations, rule updates).

Primary stakeholders

- Issuer operations, compliance, finance; custodian; admin/TA (where applicable).
- Transfer and settlement providers explicitly retained in scope, if any.
- External auditors/advisers for periodic reviews and regulatory updates.

Dependencies / gates

- Stable operating model and evidence trail sufficient for audit/regulatory scrutiny.
- Clear change-management process for token rules, disclosures, and provider configuration.
- Budgeting and resourcing for ongoing operations (often underestimated in early planning).

Risk hotspots

- Compliance drift as rules change or distribution expands across jurisdictions.
- Operational fragility under low-liquidity or stress scenarios (settlement failures, disputes).
- Provider concentration risk and cost creep relative to value delivered.

Appendix B — TVAC Provider Directory (illustrative)

This appendix is an illustrative reference layer intended to help the reader think about which types of external partners may be relevant in a tokenization project.

It is not a recommendation list, procurement shortlist, or ranking. Inclusion or omission does not imply endorsement, criticism, quality judgment, regulatory approval, or case suitability. Any provider mentioned here must be assessed independently for legal fit, jurisdictional coverage, operating model fit, technical capability, conflicts, cost, and execution reliability. This appendix is non-exhaustive and subject to change.

B1. How to use this appendix

Use this appendix to identify the provider categories a case may require, not as a signal that a specific provider should be engaged.

In many cases, the main value of the directory is not the names themselves, but the discipline of asking:

- Which functions must be covered?
- Which functions can be combined?
- Which functions must remain regulated or jurisdiction-specific?
- Which dependencies create concentration risk or vendor-capture risk?
- Which parts of the stack are critical before launch, and which can wait?

B2. Typical provider categories

Depending on the case, a tokenization project may require some or all of the following partner categories:

- legal & regulatory advisers
- tax advisers
- issuance / tokenization platform
- smart-contract / token engineering support
- custody / wallet infrastructure
- transfer agent / registry administration
- fund administration or SPV administration
- trading venue / ATS / MTF / exchange partner
- transfer or liquidity-support provider, only where explicitly required and legally permitted
- KYC / AML / onboarding provider
- payment / settlement / banking partner
- audit / assurance / cybersecurity reviewers
- valuation, data, or oracle-related providers
- distribution / placement / investor onboarding partners
- systems integration / workflow automation support

B3. Illustrative provider examples

The examples below are illustrative only and are grouped by function. They are not recommendations, rankings, or evidence of suitability for this case.

B3.1 Legal & regulatory advisers

These advisers help determine whether the proposed structure is legally enforceable, properly classified, and consistent with the intended issuance, distribution, transfer, custody, and disclosure model.

Typical use cases:

- wrapper design
- securities-law classification

- cross-border marketing analysis
- transfer restriction design
- offering documentation
- legal opinions and regulatory scoping

Illustrative examples:

- CMS
- Cooley
- FIN LAW
- MME
- NÄGELE

B3.2 Issuing platforms

These providers typically support issuance workflows, lifecycle management, permissions, investor access rules, and parts of the reporting or compliance stack.

Typical use cases:

- token issuance
- lifecycle events
- permissions / whitelisting
- cap-table or investor-position management
- compliance rule integration
- reporting workflows

Illustrative examples:

- Aktionariat
- Polymesh (Polymath)
- Realiz Group
- Securitize
- Token City
- Tokeny

B3.3 Custody, trading & settlement

These providers support safekeeping, wallet infrastructure, execution, venue access, settlement, and post-trade operations.

Typical use cases:

- institutional custody
- secure wallet infrastructure
- settlement support
- exchange / venue access
- post-trade servicing
- collateral mobility or transfer control support

Illustrative examples:

- ADDX
- Anchorage Digital
- Archax
- BitGo
- Coinbase Institutional

- Copper
- Ondo Finance
- SIX Digital Exchange (SDX)
- Sygnum Bank
- Taurus

B4. Practical selection principles

When evaluating providers, the reader should typically assess:

- jurisdictional fit
- regulatory status where relevant
- experience with the relevant asset class
- operating-model compatibility
- integration burden
- exit options / portability
- fee model
- service resilience
- auditability and evidence quality
- concentration risk

Where possible, provider choices should reinforce the case logic rather than undermine it. A weak provider stack can easily erode the value thesis of an otherwise promising structure.

B5. Important caution

The right provider stack is highly case-dependent.

A provider that is suitable for a closed institutional collateral workflow may be a poor fit for a cross-border fund wrapper. A provider that is strong in digital-asset custody may not be the right choice for transfer administration or legal structuring. A venue that looks attractive commercially may create regulatory or operating-model problems if the case does not actually need that level of transferability.

Appendix C — Glossary & abbreviations

A practical reference for the most common terms and abbreviations used in this report. Spelling note: tokenization uses “z”.

Term	Meaning
Accredited investor	An investor category used primarily in the United States for persons or entities meeting specific income, net worth, or institutional eligibility thresholds for participation in certain exempt offerings.
Added Value (AV)	TVAC’s net assessment of whether tokenization appears to improve a case overall once both upside and friction are weighed together. Formula: $AV = (NO + CS + RR) - (TC + NR)$.
AML	Anti-Money Laundering controls intended to detect, prevent, and report suspicious financial activity.
Assessment input	TVAC’s qualitative indication of whether the required input sections were completed sufficiently for the assessment to run. It measures input-form completion only and does not imply factual certainty, independent validation or supporting evidence.
Asset owner / Issuer	The originating party, sponsor, manager, or issuing structure responsible for launching or sponsoring the tokenized case being assessed.
ATS	Alternative Trading System. A regulated trading venue category used in some jurisdictions, particularly the United States.
Bankruptcy-remote SPV	A special-purpose vehicle structured to reduce the risk that the insolvency of a sponsor or related party directly affects the ring-fenced assets or obligations of the issuance vehicle.
Cap table	Capitalization table. A record of ownership interests, investor positions, or security holders in a structure or company.
Cash leg	The cash side of a transaction or settlement flow, as distinct from the asset or token side.
Conditional Go	A Tokenization Value Verdict indicating that tokenization may add net value, but remains materially dependent on design discipline, implementation choices, and resolution of key gating conditions.
Counterparty risk	The risk that one party to a transaction fails to perform its contractual obligations.
CSD	Central Securities Depository.
Custody	The safeguarding, control, and recordkeeping of assets or positions on behalf of investors or market participants.
DeFi	Decentralized Finance. A blockchain-based ecosystem of protocols and applications designed to enable financial activity without relying entirely on traditional centralized intermediaries.
DEX	Decentralized Exchange.
Distribution scope	The intended investor reach of an offering or tokenized structure, including geography, investor type, and access conditions.
DvP	Delivery versus Payment. A settlement model in which transfer of the asset and transfer of payment are linked so that one does not occur without the other.
Economic rights	The financial entitlements attached to a token or instrument, such as income participation, coupon, principal repayment, or profit-sharing.

Term	Meaning
Eligibility controls	Rules and mechanisms that determine who may acquire, hold, or transfer a token or instrument.
ESG	Environmental, Social, and Governance factors.
Execution risk	The risk that a concept that appears viable in theory fails in practice because implementation, coordination, economics, or governance proves weaker than assumed.
Factor breakdown	The report section explaining how each of TVAC's five factors contributes to the overall assessment.
Friction factors	In TVAC, the two downside factors: Tokenization Costs (TC) and New Risks (NR).
Governance	The decision-making, control, accountability, and escalation framework surrounding a structure, platform, issuer, or operating model.
Institutional investor	A professional market participant such as a fund, bank, insurer, treasury function, pension investor, or other organization investing at scale.
Investor segmentation	The way the target investor base is defined and limited by investor type, sophistication, jurisdiction, or eligibility criteria.
KYC	Know Your Customer procedures used to verify customer identity and onboarding eligibility.
Legal wrapper	The legal structure through which tokenized rights or exposures are issued, held, or governed.
Liquidity window	A predefined period or mechanism allowing limited secondary trading, transfer, redemption, or exit opportunities.
MTF	Multilateral Trading Facility under EU regulatory frameworks.
Net Added Value	See Added Value (AV).
New Opportunities (NO)	Case-specific upside that tokenization creates or materially strengthens, such as new investor reach, new operational utility, collateral use, programmability, or improved strategic flexibility.
New Risks (NR)	Additional risks introduced by tokenization, including legal, regulatory, operational, technological, governance, distribution, or market-structure risks.
NO / CS / RR / TC / NR	Shorthand for TVAC's five factors: New Opportunities, Cost Savings, Risk Reduction, Tokenization Costs, and New Risks.
Non-viable	A Tokenization Value Verdict indicating that the current structure does not support a positive tokenization value case under the TVAC methodology and should not proceed without substantial redesign.
Off-chain settlement	A settlement process that occurs through conventional systems, institutions, or cash rails rather than being fully completed on a blockchain.
On-chain settlement	A settlement process in which the relevant transfer logic is executed on a blockchain or distributed ledger environment.
Operational costs	The ongoing cost burden of running a structure, including servicing, reporting, compliance, governance, custody, registry, and technology overhead.

Term	Meaning
Operational utility	Practical workflow benefit created by tokenization, such as faster settlement, better collateral mobility, reduced reconciliation, or more efficient transfer control.
Parallel processes	A situation in which tokenized and legacy workflows both remain active, limiting the real efficiency gains from tokenization.
Permissioned	A system design in which access, participation, or transfers are restricted to approved participants under defined rules.
Professional investor	An investor category used in many jurisdictions for parties assumed to have greater sophistication, capacity, or experience than retail investors.
Prospectus	A formal disclosure document that may be required for certain public offerings or investor solicitations under applicable law.
Provider Directory	A reference section listing indicative categories of service providers relevant to tokenization implementation. It is illustrative only and not a recommendation list.
Reg D / Reg S	Common shorthand for U.S. securities offering exemptions under Regulation D and Regulation S.
Registry	The official or operational record of holdings, ownership, or investor positions.
Regulatory coverage	The jurisdiction-specific scope supported by activated, versioned regulatory claims in the TVAC profile. It does not determine how the law applies to the final facts.
Regulatory perimeter	The boundary of laws, rules, approvals, and supervisory expectations that may apply to a structure or activity.
Retail investor	A non-professional investor who generally receives higher levels of regulatory protection than professional or institutional investors.
Rights mapping	The process of making clear what legal, economic, governance, and transfer rights a token actually represents and how those rights are enforceable.
Risk heatmap	A report visualization that summarizes the relative severity and concentration of key risk areas identified in the case.
Risk Reduction (RR)	The extent to which tokenization reduces existing risk in the conventional setup, for example through better controls, improved transparency, clearer auditability, or lower reconciliation burden.
Self-custody	A model in which the end holder directly controls the relevant wallet, keys, or token position without relying on a traditional third-party custodian.
Settlement model	The way asset transfer, cash transfer, record updates, and finality are operationally achieved.
SPV	Special Purpose Vehicle. A ring-fenced legal entity often used for issuance, structuring, asset isolation, or liability segregation.
Stakeholder added value	A TVAC perspective layer showing how value and friction may be distributed differently across key participants such as the issuer, investors, and infrastructure providers.
Stop criteria	Standard decision-integrity triggers indicating that a case should stop, re-scope, or be materially redesigned unless the issue is resolved.
Structured product	A financial instrument combining defined economic exposure with a formal legal wrapper and specified payout logic.

Term	Meaning
Supporting evidence	TVAC's qualitative indicator of how far critical legal, commercial, provider and operational assumptions are supported by evidence described by the user. User-reported evidence is not independently verified by TVAC, and supporting evidence is not required for an initial assessment.
TA	Transfer Agent.
Tokenization Costs (TC)	The additional costs introduced by tokenization, including structuring, legal work, compliance, integration, platform costs, governance overhead, servicing changes, and operating complexity.
Transfer controls	Rules, permissions, and technical or legal mechanisms governing who may transfer, receive, or hold the token.
Transfer restrictions	Specific limitations placed on secondary transfers, often for regulatory, contractual, or investor-eligibility reasons.
TVAC	Tokenization Value-Add Calculator. A structured decision-support methodology for assessing whether tokenization appears to create net added value in a specific case.
Value pathways	The concrete ways in which tokenization might create value in the case, such as improved distribution, lower friction, stronger control, faster settlement, or better collateral use.
Vendor-capture risk	The risk that a structure becomes economically or operationally overly dependent on one platform, provider, or implementation path in a way that shifts value away from the initiating party.
Venue fit	The degree to which the chosen issuance, transfer, trading, or liquidity venue matches the asset, investor scope, regulatory posture, and operating model of the case.
Verified regulatory claim	A versioned statement derived from an identified source and approved for conditional regulatory scoping. It is not a legal opinion or case-specific compliance conclusion.
Viable	A Tokenization Value Verdict indicating that the TVAC methodology identifies a positive tokenization value case under the described structure, subject to validation and disciplined execution. It is not an investment, credit, yield, valuation, legal, tax, or suitability assessment.
Whitelist / Whitelisting	A mechanism for restricting participation or transferability to pre-approved investors, wallets, counterparties, or participants.

Appendix D — User-supplied case inputs

D.1 Full verbatim case description

Full original free-text case input provided by the user.

We are a US-based corporate issuer with an existing, recurring commercial paper (CP) program used for short-term working capital. Today the program relies on multiple intermediaries, manual confirmations, repeated reconciliations between the issuer's treasury team, issuing/placing agents, custodians, and investors, and a settlement cycle that creates operational friction and occasional settlement fails.

We want to introduce a tokenized CP note format (economically equivalent to our existing CP) for professional and accredited investors only. This is not a retail product and not a public exchange initiative. The scope is deliberately conservative: a single US issuance + distribution perimeter, whitelisted participation, and institutional-grade custody. The primary objective is post-trade efficiency and risk reduction rather than “new liquidity”.

The tokenized format is designed to (1) reduce reconciliation breaks by moving to a single shared position record, (2) automate issuance and lifecycle events (issuance confirmations, coupon/accrual reporting, maturity repayment notices), and (3) tighten settlement controls through delivery-versus-payment (DvP) style workflows. Importantly, we commit to decommissioning duplicated legacy steps for the tokenized tranche (no permanent parallel “double-booking”), so savings are structural rather than cosmetic.

We expect measurable cost savings (reduced agent/custody operations per issuance, fewer manual exceptions) and measurable risk reduction (fewer settlement fails, cleaner audit trail, reduced counterparty/operational exposure). Any secondary transfers, if permitted at all, are tightly controlled and limited to eligible whitelisted holders under standard securities restrictions. The issuer is the primary beneficiary of the operational savings and control improvements under this design.

D.2 Structured input record

This record reproduces the structured inputs supplied for this assessment. For dropdowns and checkboxes, it also shows the alternatives that were available but not selected.

Field	User response	Other available options not selected
Issuance jurisdiction	United States	Not applicable
Sale jurisdictions (comma-separated)	United States	Not applicable
Instrument type	Digital wrapper (CSD/TA/registrar)	Sole-record / native tokenized security
Custody model	Regulated custodian	Transfer agent / CSD; Self-custody (wallets)
Secondary-market liquidity	None	Ad-hoc OTC; Quarterly windows; Continuous
Secondary-market venue	OTC	ATS / MTF; DEX (permissioned); CSD / TA-registered
Payment / cash-leg model	T+1/T+2 securities settlement	Not decided yet; Not material at this stage; Traditional bank / fiat payments; Custodian / bank / TA / CSD settlement; Stablecoin / tokenized cash; Deposit token / tokenized deposit / institutional DvP rail; Advanced: T+0 / atomic DvP; Advanced: off-chain booking
Offering disclosure	Exempt / limited-offering route (jurisdiction-dependent)	KID (PRIIPs); Prospectus
Transfer restrictions	Whitelist + accredited/professional-only + issuer/agent-approved transfers	Not applicable
Tax considerations (optional)	Standard US withholding and investor reporting depend on holder type; tax handled under existing CP program processes.	Not applicable
Investor types	Professional; Qualified / accredited (jurisdiction-dependent)	Retail

An option shown as not selected was not necessarily actively considered or rejected by the user; it was simply not selected for this assessment.

Appendix E — Disclaimer

Disclaimer

This report is provided for informational and decision-support purposes only. It does not constitute legal, regulatory, tax, accounting, investment, valuation, or financial advice, and it should not be relied upon as a substitute for professional advice tailored to the specific facts and circumstances of the case.

TVAC is a structured assessment tool designed to evaluate whether and when tokenization may create added value relative to conventional alternatives. Its outputs reflect a methodology-based analytical framework operating under uncertainty. They are not guarantees of commercial success, feasibility, investor demand, regulatory approval, liquidity, execution quality, budget outcome, or risk reduction.

Any scores, verdicts, scenarios, stakeholder assessments, suggested actions, implementation observations, and roadmap elements in this report are indicative only and depend on the quality, completeness, and interpretation of the input data, as well as on assumptions about structure, jurisdiction, service-provider readiness, market conditions, execution capability, and timing.

Tokenization outcomes can differ materially from this assessment if key conditions change, including legal structuring, investor scope, jurisdiction count, custody model, distribution pathway, settlement design, governance arrangements, servicing readiness, technology integration, cost allocation, evidence quality, or liquidity support.

A TVAC verdict is a tokenization added-value verdict only. It must not be presented or used as investment approval, credit rating, yield validation, issuer endorsement, regulatory approval, due diligence report, or investor recommendation. TVAC does not validate projected returns, coupon payments, credit quality, valuations, legal enforceability, tax treatment, or investor suitability.

This report does not replace formal legal analysis, regulatory review, tax analysis, technical architecture review, cybersecurity review, accounting assessment, operational diligence, commercial validation, or board-level decision processes. Independent advice should always be obtained before any structuring, issuance, distribution, investment, operational, or implementation decision is made.

Neither TVAC nor Astoval assumes responsibility for actions taken on the basis of this report alone. Users remain solely responsible for all commercial, legal, regulatory, operational, and strategic decisions related to any proposed tokenization initiative.



Tokenization Value-Add Calculator

A PRODUCT BY ASTOVAL

Astoval provides independent, AI-enhanced early-stage decision support for asset tokenization, helping organizations assess whether tokenization creates genuine added value before significant resources are committed to legal structuring, platform selection or implementation.

PRIVACY AND DATA

Case inputs and generated reports are processed for delivery and are not intentionally retained as a persistent customer-case record in the standard self-service flow. OpenAI API data is not used for model training unless the API customer opts in; separate abuse-monitoring retention may apply for up to 30 days.

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